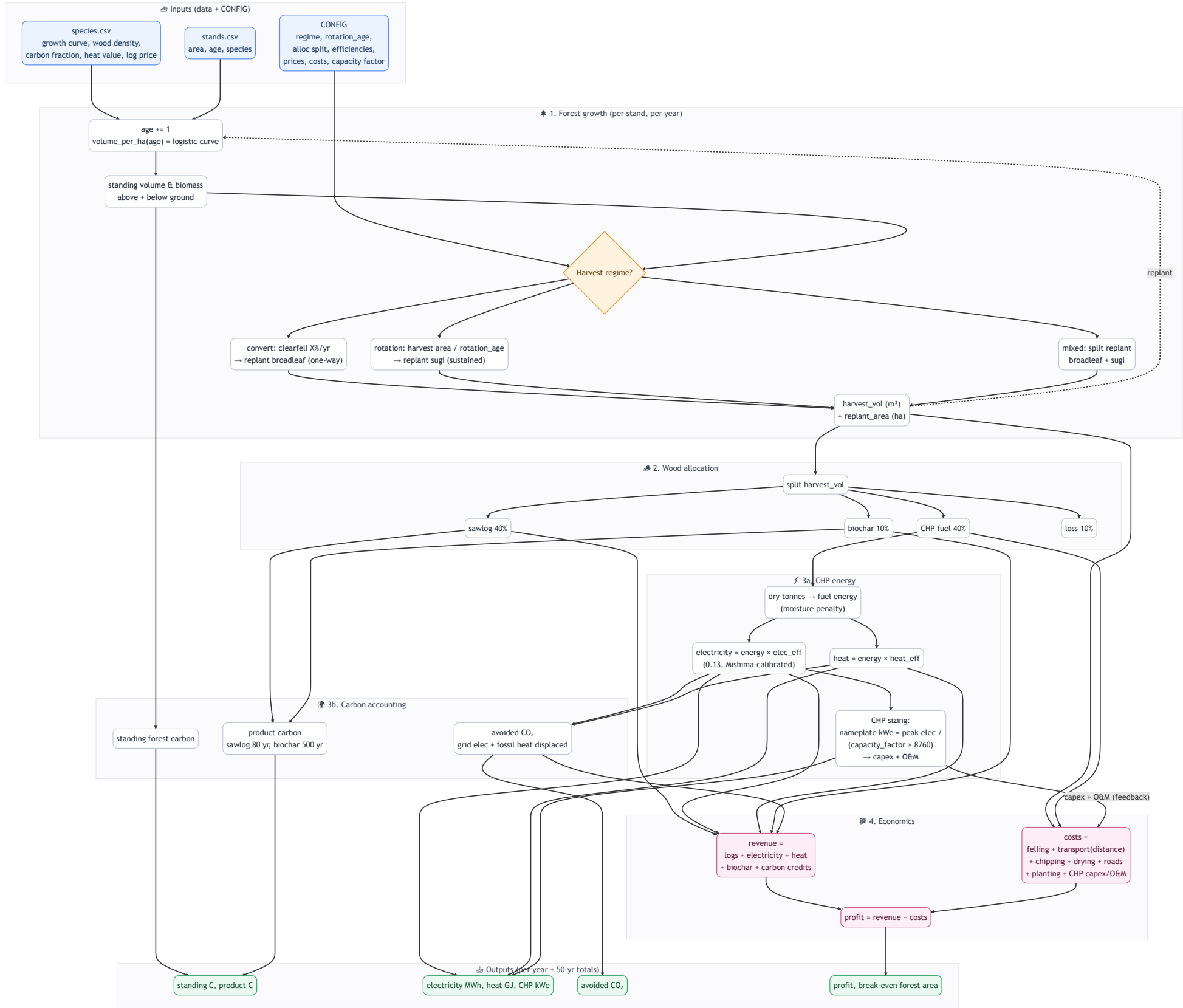


Forest Twin — model interaction diagram

How the variables flow through the simulator, one year at a time. Inputs (blue) → the three loops (forest / energy+carbon / money, pink) → outputs (green). The CHP is **auto-sized** from the fuel flow, so capex/O&M feed back into the economics.



The one-line story

`species + stands + CONFIG` → grow the forest → harvest it (the **regime** decides if fuel is sustainable) → split the wood → burn some for **power + heat** and lock the rest as **stored carbon** → tally **avoided CO₂** → subtract real supply-chain **costs** (sized to a right-fit CHP) → **profit** and the **forest-area needed** for a given CHP size.

The single most important link is the dotted **replant** → **grow** arrow: in *rotation* mode it closes the loop (perpetual fuel); in *convert* mode it doesn't (broadleaf is never re-harvested → fuel runs out).